

MID-STATE TECHNICAL COLLEGE • IT PROGRAMS

Software Defined Data Center

Unit 1 • Virtualization (10-150-130)



Learning Outcomes

By the end of this unit, you will be able to:



Explain the software defined data center

Describe how compute, storage, network, and security are abstracted and delivered as services.



Classify virtualization technologies

Distinguish server, desktop, application, network, and storage virtualization by what each abstracts.



Compare hypervisor architectures

Contrast Type 1 and Type 2 designs and justify a platform choice for a given scenario.



Reason about resource allocation

Apply CPU, memory, storage, and network allocation concepts to size a virtual machine.

What Is a Software Defined Data Center?

An SDDC abstracts every layer of infrastructure into software-defined, policy-driven services.

- All four pillars — **compute**, **storage**, **networking**, and **security** — are virtualized and consumed as services.
- Hardware becomes a **resource pool** rather than a set of individually configured boxes.
- Provisioning is driven by **policy and automation**, not by a technician at a rack.
- Capacity is allocated to workloads dynamically, so utilization rises without new purchase orders.



The traditional contrast

A legacy data center ties one application to one server, sized for peak load and idle the rest of the time. Change means a purchase, a rack visit, and a maintenance window.

Why This Matters in 2026

Platform choice used to be automatic. It is now one of the most consequential decisions in IT.

- Broadcom moved VMware to **subscription-only, per-core licensing** with per-CPU core minimums; perpetual licenses are gone.
- Small and mid-size environments have seen renewal costs rise several times over for the same hardware.
- Gartner expects roughly **35% of today's VMware workloads** to be running elsewhere by 2028.
- Result: **Proxmox VE** and **Hyper-V** are now mainstream destinations — and the two platforms this course teaches.

Career framing: Employers are hiring for hypervisor-agnostic skills. Knowing why a platform fits matters more than knowing one vendor's menus.

Market conditions current as of mid-2026; verify pricing details against vendor sources before quoting them.

LESSON 1

Accessing the Lab Environment



The MSTC POD Architecture

Every student is assigned a physical POD in one of two Mid-State datacenters.

- Compute is an **HP DL20 Gen 10** server running Proxmox VE as a bare-metal hypervisor.
- Shared storage is a **TrueNAS SCALE** node presenting NFS and iSCSI to the POD.
- Two sites: **Wisconsin Rapids** (prefix 22) and **Stevens Point** (prefix 31).
- Installation media lives on an NFS share at `10.xx.2.254:/ISOs`.



Physical, not simulated

Your POD is hardware someone can trip over. Power states, firmware, and cabling are all in scope — which is exactly the point.

Three Ways In

Each access path solves a different problem — know which one to reach for.

Path	What it reaches	When you need it	Layer
OpenVPN	The lab network itself	Any off-campus work	Network
HP iLO5	The server, powered off	Install, BIOS, no-boot	Out-of-band
Proxmox web UI	The running hypervisor	Daily VM management	In-band
SSH / console	The host shell	CLI tasks, scripting	In-band

Out-of-band management is the reason you can rebuild a host you cannot log into. It is not a convenience feature.

Naming and Addressing Conventions

The lab uses a structured scheme so any address tells you where you are.

```
POD name    WRPOD01          SPPOD03
Host FQDN   22POD01-Host1.mstclab.local
Address     10.xx.3.yy      xx = site (22 = WR, 31 = SP)  yy = POD host
```

- The **site prefix** is the second octet — 22 in Wisconsin Rapids, 31 in Stevens Point.
- The **third octet** separates management (.2.) from POD traffic (.3.).
- Your POD assignment supplies yy — the host portion. It does not change during the term.

Try one: 10.31.3.42 — which building is that host in, and is it on the management network or the POD network?

Working Safely in a Shared Lab

- Lab credentials are standardized (`root`, `truenas_admin`) so instruction can move quickly.
- Document every change you make; an undocumented change is indistinguishable from a fault.
- Snapshot before you experiment — rollback is the cheapest recovery you will ever have.
- Stay inside your own POD. Neighboring PODs are other students' graded work.



SECURITY FOCUS

Shared, well-known credentials are acceptable in a teaching lab and nowhere else. In production, hypervisor management accounts are the highest-value target on the network.

Why it matters: Compromise the hypervisor and every virtual machine on it is compromised at once.

LESSON 2

Virtualization Fundamentals



From Mainframe to Cloud

Virtualization is over sixty years old. Only the hardware it runs on is new.

Era	Development	Problem it solved
1960s	IBM CP-40 and CP-67 mainframe partitioning	One costly machine, many users
1999-2001	x86 virtualization arrives on commodity servers	Sprawling, idle rack servers
2005-2006	Intel VT-x and AMD-V hardware assist	Software emulation was slow
2013-present	Containers, then Kubernetes orchestration	Full OS per app was heavy

Each wave abstracted one more layer away from the hardware — and none of them replaced the one before it.

The Problem Virtualization Solves

One application per physical server means most of your hardware is idle most of the time.

- Dedicated servers commonly ran at **10-15% CPU utilization** while drawing full power.
- Consolidation onto a hypervisor typically raises that to **70-80%** on the same hardware.
- Fewer physical hosts means less power, cooling, floor space, and fewer support contracts.
- Provisioning drops from weeks of procurement to minutes of allocation.



Why so idle?

Every server was sized for its worst hour — quarter close, backup window, month-end batch. The other 8,700 hours it coasted.

Business Drivers

Financial

- Fewer hosts to buy, power, and cool
- Capital spend shifts toward operating spend
- Licensing consolidates onto fewer sockets or cores
- Deferred data center expansion

Operational

- Snapshots and replication make recovery routine
- Live migration removes most maintenance windows
- Templates make new systems repeatable, not bespoke
- Test environments cost allocation, not procurement

The argument that wins: Virtualization is bought for agility and recovery at least as often as it is bought for savings.

Honest Limits

Virtualization is not free, and it is not always the right answer.

- **Performance overhead** is small but real; latency-sensitive and hardware-bound workloads still lose.
- **Licensing complexity** grows — vendors license by socket, core, VM, or user, and the rules change.
- **Consolidation risk**: one failed host now takes down twenty services instead of one.
- **Skills and tooling** — monitoring, backup, and troubleshooting all have to be relearned.



Know when not to

Hardware dongles, specialized PCI cards, real-time control systems, and vendor-unsupported appliances are still legitimate reasons to stay physical.

LESSON 3

Types of Virtualization



Five Layers, One Idea

Each type abstracts a different resource from the hardware underneath it.

Type	What it abstracts	Typical example
Server	The whole machine	Proxmox VE, Hyper-V, ESXi
Desktop	The user workstation	VDI, remote desktop pools
Application	The app from the OS	Containers, app streaming
Network	Topology from cabling	SDN, VLANs, overlays
Storage	Capacity from devices	SANs, ZFS pools, Ceph

Real infrastructure stacks these — a VM on virtual storage reached through a virtual network.

Server Virtualization

The foundation layer — many virtual machines sharing one physical host.

- A **virtual machine** is a configuration file plus one or more virtual disks — nothing more.
- Because it is files, a VM can be **snapshotted**, cloned, backed up, and migrated while running.
- The hypervisor schedules real CPU time and memory pages among the guests.
- Guests are **isolated**: a crashed VM does not take its neighbors with it.



Hardware independence

A VM sees generic virtual devices, not your actual NIC or controller. That is what lets it move to different hardware unchanged.

Desktop and Application Virtualization

Desktop (VDI)

- Desktop operating systems run in the datacenter; users connect remotely
- **Persistent** desktops keep user changes between sessions
- **Non-persistent** desktops reset to a golden image at logoff
- Data never leaves the datacenter, which simplifies BYOD and offboarding

Application

- The application is isolated from the operating system beneath it
- Multiple versions of the same software coexist without conflict
- **Containers** package an app with its dependencies, not a whole OS
- Streaming and just-in-time delivery avoid full installs

Both types move the management burden from the endpoint to the datacenter — where you already have staff and backups.

Network and Storage Virtualization

- **Software Defined Networking** separates the control plane from the forwarding hardware.
- Virtual switches, VLANs, and overlays create topologies with no cabling changes.
- **Microsegmentation** applies firewall policy between individual workloads, not just at the perimeter.
- **Storage virtualization** pools devices into one logical capacity — with thin provisioning and live migration.



SECURITY FOCUS

Flat networks let an attacker who compromises one server reach every other server. Microsegmentation confines that first foothold to a single workload.

Why it matters: Lateral movement, not initial access, is what turns an incident into a breach.

Containers vs. Virtual Machines

Both isolate workloads. They draw the isolation boundary in completely different places.

	Virtual machine	Container
Includes	A full guest operating system	App and dependencies only
Kernel	Its own	Shared with the host
Start time	Seconds to minutes	Milliseconds
Isolation	Hardware-enforced, strong	Kernel-enforced, weaker
Good for	Mixed OSes, legacy apps	Scale-out, stateless services

Proxmox VE runs both: KVM virtual machines and LXC containers, managed from the same interface.

LESSON 4

Hypervisor Architecture



What a Hypervisor Actually Does

The hypervisor owns the physical hardware and arbitrates every guest's access to it.

- Schedules physical CPU time across all guests and enforces their limits.
- Maps guest memory pages to real memory and keeps guests out of each other's address space.
- Presents **virtual devices** — disk, NIC, display — that guests treat as ordinary hardware.
- Relies on **hardware assist** — Intel VT-x, AMD-V, and SLAT — to do all of this at near-native speed.



Check the firmware

If virtualization extensions are disabled in BIOS, the hypervisor either refuses to install or runs unusably slow. It is the first thing to verify on new hardware.

Type 1 vs. Type 2

	Type 1 — bare metal	Type 2 — hosted
Runs on	The hardware directly	A general-purpose OS
Performance	Near native	Host OS takes a share
Attack surface	Small and purpose-built	Whole host OS included
Typical use	Production datacenters	Development, training, testing
Examples	Proxmox VE, Hyper-V, ESXi	VirtualBox, VMware Workstation

The question that decides it: Is there a general-purpose operating system between the hypervisor and the hardware? If yes, it is Type 2.

Proxmox VE is Type 1: KVM lives in the Linux kernel itself, with Debian supplying only the management surface.

Type 1 Platforms in 2026

The production hypervisor market as students will encounter it this year.

Platform	Foundation	Licensing model
Proxmox VE 9.x	Debian 13, KVM and LXC	Open source; optional support tier
Hyper-V (Server 2025)	Windows Server role	Included with Windows Server
VMware vSphere	Purpose-built ESXi	Subscription, per core
Nutanix AHV	KVM-based, hyperconverged	Subscription, per node
XCP-ng	Xen-based, community	Open source; optional support

Versions current as of August 2026 — confirm the release your employer actually runs before you plan against it.

Proxmox VE at a Glance

Your platform for Units 2 through 12 — installed on your own POD next week.

- Type 1 hypervisor built on **Debian Linux** with **KVM** for VMs and **LXC** for containers.
- Clustering, live migration, and high availability are included — no feature is behind a license tier.
- Storage flexibility: local, ZFS, NFS, iSCSI, and Ceph, all managed from one web interface.
- Everything in the UI is also available through the `pvesh` CLI and a REST API.



Current release

The 9.x line is built on Debian 13. Recent releases added SDN fabrics, LVM snapshot support for SAN storage, and a dynamic cluster load balancer.

LIVE DEMO

Log into a Proxmox VE node and walk the datacenter tree, then the node summary.

Hyper-V at a Glance

Your platform for Units 13 through 16 — a Type 1 hypervisor delivered as a Windows Server role.

- Enabling the role inserts the hypervisor **beneath** Windows; the host OS becomes a privileged partition.
- Windows Server 2025 raised Generation 2 limits to **2,048 vCPUs** and 240 TB of memory per VM.
- **GPU partitioning** now shares one physical GPU across VMs and still permits live migration.
- **Workgroup clusters** allow failover clustering without Active Directory — useful at the edge.



Why both platforms?

Employers run mixed estates. Learning one open-source and one Microsoft hypervisor makes the concepts portable instead of vendor-specific.

Security in a Shared Data Center

Consolidation concentrates risk. The management plane is now the crown jewel.



SECURITY FOCUS

Anyone with hypervisor administrator rights can read the disk of every guest, clone it, or power it off — regardless of the guest's own access controls.

Isolation between guests is enforced by the hypervisor, so hypervisor vulnerabilities and misconfigurations are the ones that matter most. Keep the management network separate from guest traffic, patch the host on its own schedule, and treat administrator accounts as privileged access, not as convenience logins.

Why it matters: A guest OS cannot defend itself against the layer it runs on.

LESSON 5

Resource Allocation



Virtual vs. Physical Resource Models

Physical model

- Resources bought up front and sized for peak load
- Idle capacity is stranded in the box that owns it
- Changing the sizing means new hardware
- Utilization commonly in the low teens

Virtual model

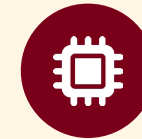
- Resources drawn from a shared pool on demand
- **Oversubscription** works because peaks rarely coincide
- Sizing is a setting, revisable in a maintenance window
- Utilization commonly 70-80%

The trade: Oversubscription buys efficiency and pays for it in contention. Monitoring is what tells you which side you are on.

CPU Allocation

A vCPU is a scheduling entitlement, not a piece of silicon you own.

- Guests are scheduled onto physical cores; a **vCPU** is a claim on time, not a dedicated core.
- **Shares, reservations, and limits** set priority, a guaranteed floor, and a hard ceiling.
- Over-allocating vCPUs makes a VM **slower** — it must wait for that many cores to free at once.
- Watch **CPU ready time** (host side) and **steal time** (guest side) for contention.



Start small

Size for observed load, not for the request. Adding a vCPU later is trivial; recovering from datacenter-wide contention is not.

Memory Allocation

Memory is usually the first resource to run out — and the least graceful about it.

- **Ballooning** lets the host reclaim memory a guest is not actively using.
- **Page sharing** (KSM) deduplicates identical memory pages across similar guests.
- Overcommitment is safe only while the guests' real working sets fit in physical RAM.
- When the host starts swapping, performance collapses rather than degrading gently.



CPU vs. memory

CPU contention makes everything a little slower. Memory exhaustion makes something stop. Plan the two differently.

Storage and Network Allocation

Capacity is easy to plan. Throughput is what actually bites.

Control	What it does	What to watch
Thick provisioning	Reserves full capacity up front	Wasted, unusable free space
Thin provisioning	Consumes space as data is written	Datastore filling unnoticed
IOPS limits	Caps a guest's storage operations	One VM starving the others
Network QoS	Prioritizes or caps guest bandwidth	Backups swamping production

A thin-provisioned datastore that fills up will pause every virtual machine on it simultaneously. Alert on the datastore.

Live Demo Inventory

Demonstrated live from an instructor POD — follow along, then repeat it on your own.

- Connect through OpenVPN and confirm reachability of the POD subnet.
- Open an HP iLO5 session and show power state and remote console on a host that is off.
- Log into the Proxmox VE web interface and tour the datacenter, node, and VM panes.
- Decode a lab address and hostname together, octet by octet.
- Show a running VM's configuration file and identify the virtual devices.



Bring your POD assignment sheet — you will need your host address for the lab that follows.

Hands-On Lab: Unit 1

Physical POD lab · Wisconsin Rapids and Stevens Point datacenters · your assigned POD

- Install and configure the OpenVPN client, and verify access to the lab network.
- Locate your POD assignment and record your host name and IP address.
- Establish an HP iLO5 session to your assigned server and confirm power control.
- Identify the components of your POD and diagram how they connect.
- Document every credential and address you will need for the rest of the term.



Graded on a working connection to your own POD — not on a screenshot of someone else's.

Key Takeaways



An SDDC virtualizes compute, storage, network, and security, and delivers them as policy-driven services.



Out-of-band management reaches a server the operating system cannot.



Five types of virtualization, one question: what is being abstracted from what?



Type 1 runs on the hardware; Type 2 runs on an operating system that runs on the hardware.



A vCPU is an entitlement to time — over-allocating one makes the guest slower, not faster.

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Questions?

Unit 1 · Software Defined Data Center